RETENTION RISK ANALYTICS / PORTFOLIO PROJECT

BANK CUSTOMER CHURN ANALYSIS

An evidence-led view of customer exit patterns, priority segments and retention actions

Management question: Which customer segments show elevated historical churn, and where should the bank investigate before designing targeted retention interventions?			
10,000	**2,038**	**20.38%**	**3**
Customer records	Customers exited	Historical churn rate	Geographies

Executive purpose

This project examines the distribution of historical customer churn across geography, age, product holding, engagement, account balance and selected customer attributes. It translates observed differences into a disciplined investigation and retention-testing agenda.

What this report is - and is not

It is a decision-support analysis based on a supplied 10,000-row CSV and a reproducible recalculation of key EDA metrics. It is not proof of causal drivers, a validated real-time early-warning system, or a quantified financial business case.

Prepared from the supplied raw CSV, Python EDA notebook, MySQL analysis script, business-problem brief and Power BI screenshot. Source files reviewed on 02 November 2026.

1. Executive summary

The supplied customer file contains 10,000 records, of which 2,038 have Exited = 1. The historical churn rate is 20.38%. Recalculated segment results show that churn is concentrated unevenly across geography, age, number of products, engagement and balance status.

Signal	Verified result	Decision implication
Geography	Germany 32.44% (814/2,509), versus 16.67% Spain and 16.17% France.	Run a Germany-focused diagnostic; control for customer mix before attributing the gap.
Age	50-59: 56.04% (487/869); 40-49: 30.83% (807/2,618).	Review service fit and product needs for older cohorts; do not assume age is causal.
Product count	3 products: 82.71% (n=266); 4 products: 100.00% (n=60).	Audit the multi-product journey; small segments require caution.
Engagement	Inactive: 26.87% (1,303/4,849); active: 14.27% (735/5,151).	Design a controlled re-engagement pilot and measure incremental retention.
Balance	Has balance: 24.10%; zero balance: 13.82%.	Examine balance and engagement together; avoid causal interpretation.

Business problem

The bank lacks a clear, evidence-led view of which customer profiles and engagement patterns are associated with account exit. It needs to convert historical customer data into actionable retention segments and assess where a baseline risk-classification approach could help prioritize investigation.

Project objectives

- Quantify the historical churn baseline and define the denominator consistently.
- Compare churn rates across customer and account segments, showing the segment size beside each rate.
- Use statistical tests to distinguish stronger observed associations from descriptive variation.
- Flag leakage, timing and small-sample risks before turning a segment pattern into a retention policy.
- Translate findings into diagnostics, controlled tests and a measurable monitoring plan.

Core interpretation: the analysis identifies where to investigate first. It does not establish why customers left or prove that a specific intervention will reduce churn.

2. Dataset, scope and data quality

Attribute	Observed result
Source	Supplied Kaggle Bank Customer Churn CSV
Dimensions	10,000 rows × 18 columns
Target	Exited: 1 = churned / exited; 0 = retained
Geographies	France, Germany and Spain
Unique customer check	10,000 distinct CustomerId values; no duplicate IDs
Missing values	0 across all 18 columns
Exact duplicate rows	0 in the raw file; 0 after excluding identifier/name fields
Time coverage	No transaction date, event date or observation timestamp is supplied

Fields and analytical scope

The source includes identifiers (RowNumber, CustomerId, Surname), customer attributes (CreditScore, Geography, Gender, Age, Tenure), account and engagement fields (Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary), the outcome Exited, and supplementary fields (Complain, Satisfaction Score, Card Type, Point Earned).

Row identifiers and surname are not analytical predictors. Complain is retained only for a data-quality/leakage check because timing is unknown and it is almost identical to the churn outcome. Satisfaction Score is descriptive pending confirmation of survey timing and coverage.

Preparation and validation

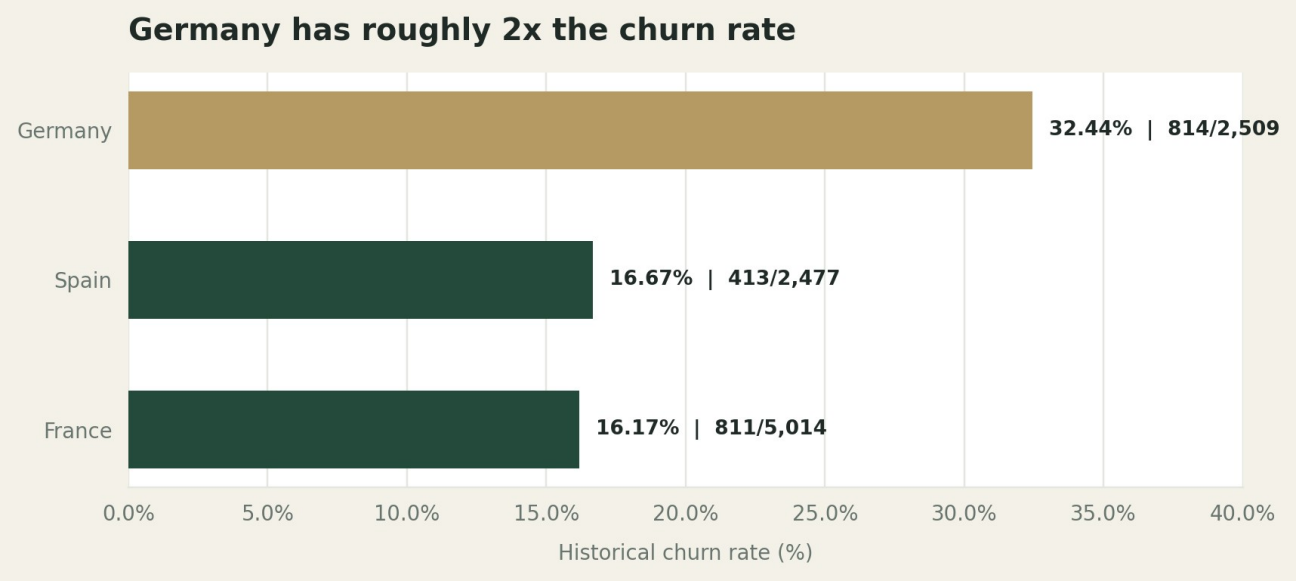
- Loaded the supplied raw CSV directly and checked dimensions, column names and data types.
- Verified target counts, uniqueness of CustomerId, missingness and exact duplicate rows.
- Created report-level age bands and balance/membership segments using explicit, documented rules.
- Recomputed group counts, churn counts and churn rates using each segment's own customer count as denominator.
- Ran chi-square tests for categorical associations and Mann-Whitney U tests for continuous age and balance differences.

Workflow

Step	Work performed	Evidence / output
1. Data audit	Profile file, target, identifiers and completeness	10k × 18; 0 missing; 0 duplicates
2. EDA	Segment churn rates with denominators	Geography, age, product, activity, balance, gender, card, satisfaction
3. Statistics	Chi-square and Mann-Whitney tests	p-values and association checks
4. SQL review	Static review of MySQL query logic	No live SQL connection or executed query results supplied
5. BI review	Inspect supplied Power BI screenshot and DAX setup notes	Visual accuracy, labels, slicer coverage and presentation QA
No date field exists in the source. Calendar trends, customer tenure-over-time curves, cohort retention and 30/60/90-day forecast horizons cannot be calculated from this dataset.		

3. Where churn concentrates: geography and age

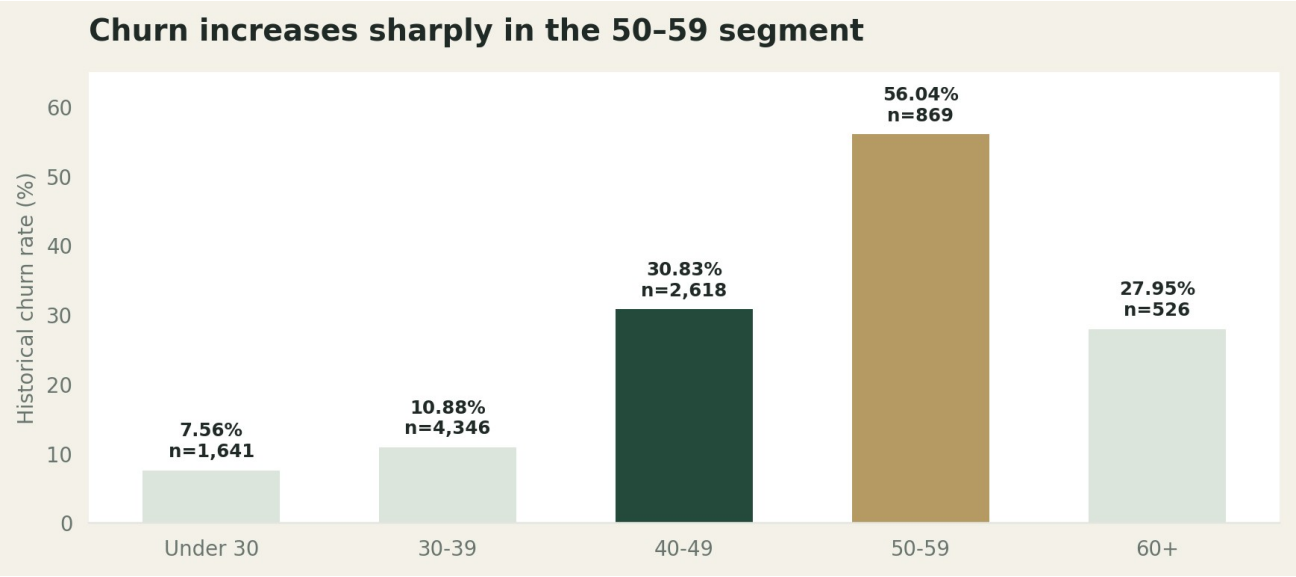
Germany is the clearest geographic outlier



Germany records 32.44% historical churn (814 of 2,509 customers), compared with 16.67% in Spain (413 of 2,477) and 16.17% in France (811 of 5,014). Germany's churned-customer count is close to France's despite having approximately half the customer base. This makes the rate comparison more revealing than raw exit counts.

A focused Germany review is justified. The next analysis should compare German customers by age, tenure, product count, balance and activity, and then control for customer mix. Relationship-manager feedback and competitor context are hypotheses to test, not conclusions from this CSV.

Churn is highest in the 50-59 age band

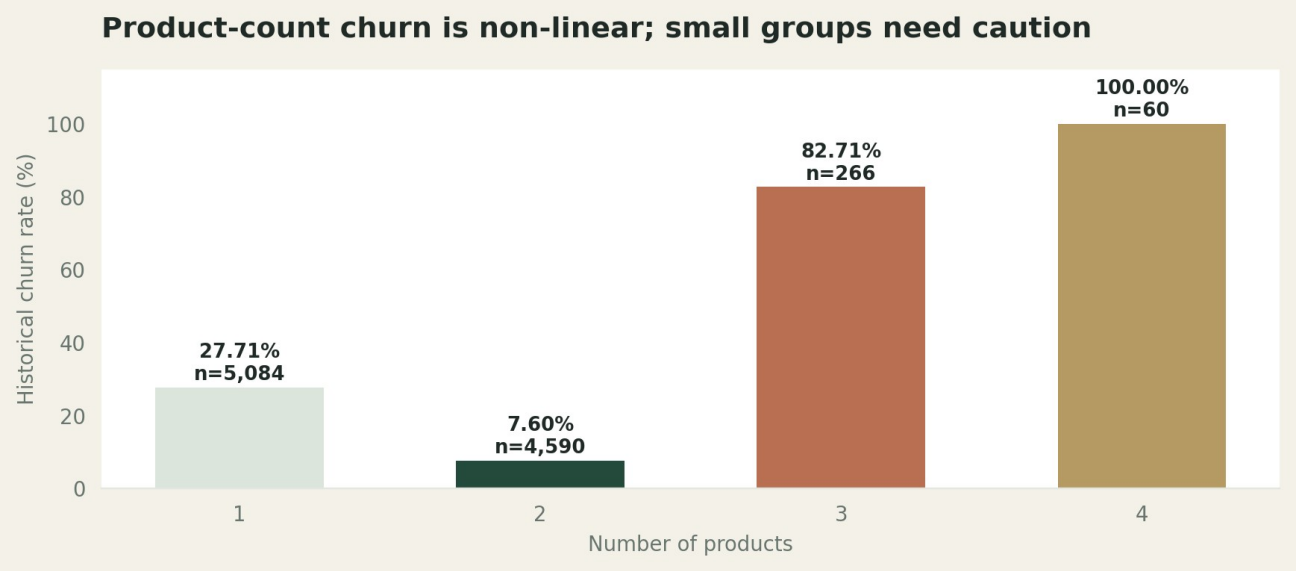


The 50-59 band has a 56.04% churn rate (487/869); the 40-49 band is 30.83% (807/2,618). Customers under 30 show 7.56% churn (124/1,641), while the 60+ segment shows 27.95% (147/526). Average age is 44.84 years among churned customers and 37.41 among retained customers.

Age patterns are statistically associated with churn in this sample (chi-square $p < 0.001$; Mann-Whitney comparison of continuous age $p < 0.001$). These results do not show that age itself causes account exit.

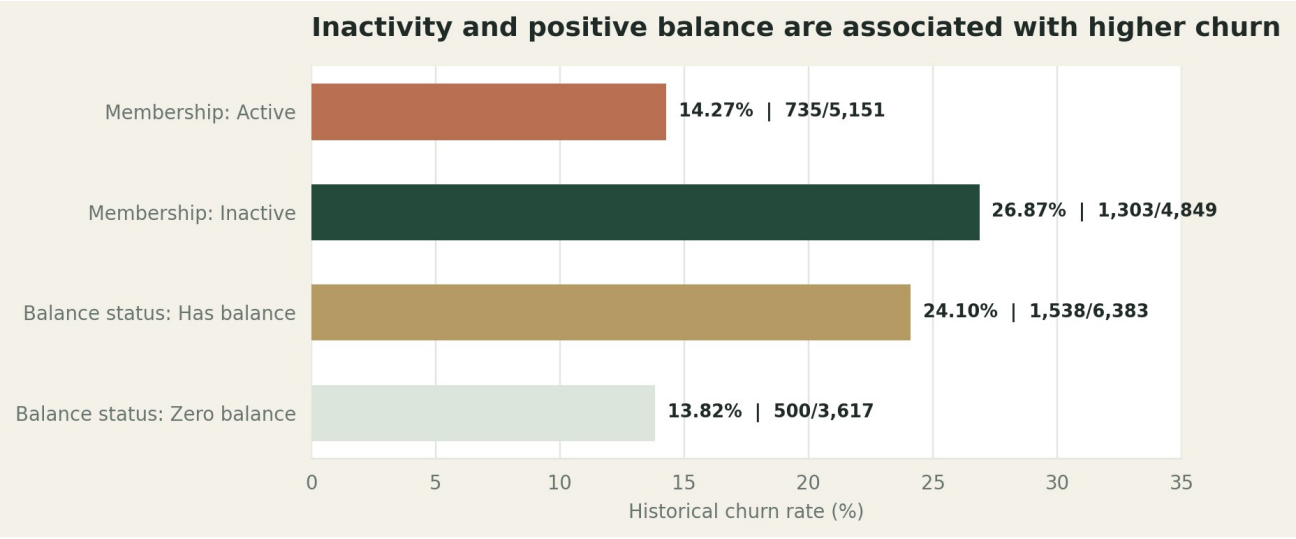
4. Product, engagement and account patterns

Product-count results are non-linear



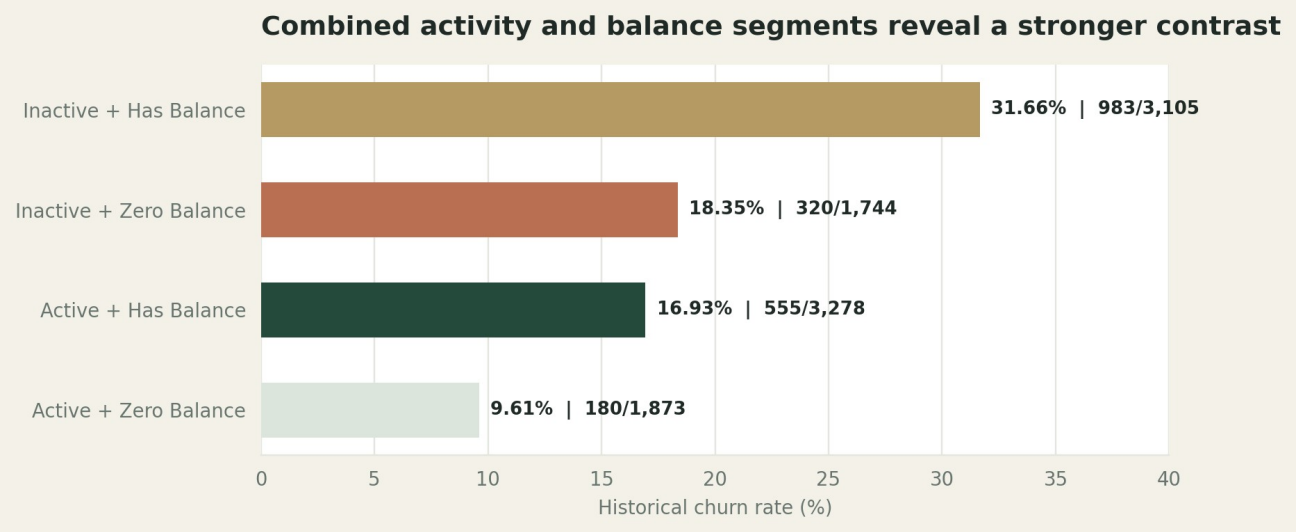
The two-product segment has the lowest churn rate at 7.60% (349/4,590). One-product customers show 27.71% churn (1,409/5,084). The three-product segment is 82.71% (220/266), while all 60 customers with four products exited in this file. The 3- and 4-product groups are small relative to the full population; validate stability and inspect customer journeys before changing product policy.

Inactive customers show a higher historical churn rate



Inactive customers have 26.87% churn (1,303/4,849) versus 14.27% among active members (735/5,151). A controlled re-engagement test is more defensible than a blanket campaign: compare contacted customers with a comparable control group and track incremental retention.

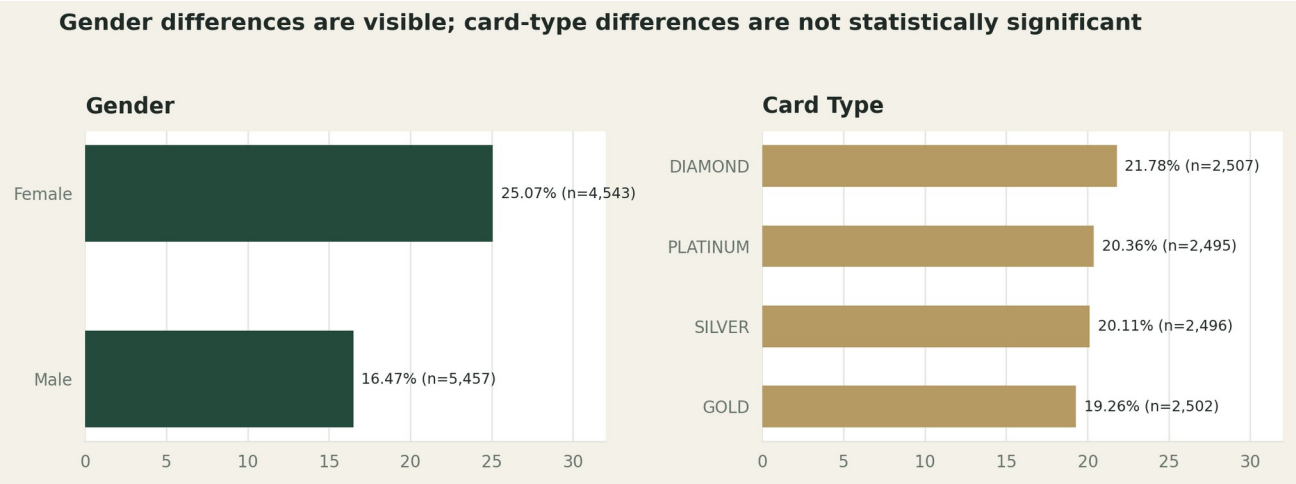
Combined balance and activity segments sharpen prioritization



Inactive customers with a positive balance show 31.66% churn (983/3,105), compared with 9.61% for active customers with zero balance (180/1,873). This combined segment view can guide investigation, but it remains observational and should be checked against customer mix.

Combined segment	Customers	Churned	Churn rate
Inactive + Has Balance	3,105	983	31.66%
Inactive + Zero Balance	1,744	320	18.35%
Active + Has Balance	3,278	555	16.93%
Active + Zero Balance	1,873	180	9.61%

5. Additional segmentation and statistical evidence



Female customers show 25.07% churn (1,139/4,543), compared with 16.47% among male customers (899/5,457). The gender association is statistically significant (chi-square $p < 0.001$), but the analysis does not explain the mechanism. Review service experience, product mix and cohort composition before considering any segment-specific intervention.

Card-type rates range from 19.26% for Gold to 21.78% for Diamond, but the difference is not statistically significant in this sample (chi-square $p = 0.168$). Satisfaction-score rates range from 19.64% to 21.80%; no significant association is detected ($p = 0.433$). Avoid describing card type or satisfaction score as churn drivers.

Statistical tests

Dimension	Test	Result	Interpretation
Geography	Chi-square	$p < 0.001$; Cramer's V = 0.173	Significant association
Age group	Chi-square	$p < 0.001$; V = 0.358	Strongest categorical association among these segments
Number of products	Chi-square	$p < 0.001$; V = 0.387	Strong association; 3-4 product groups are small
Active membership	Chi-square	$p < 0.001$; V = 0.156	Significant association
Balance group	Chi-square	$p < 0.001$; V = 0.122	Significant association
Gender	Chi-square	$p < 0.001$; V = 0.106	Significant association
Card type	Chi-square	$p = 0.168$; V = 0.022	Not significant at 0.05
Satisfaction score	Chi-square	$p = 0.433$; V = 0.020	Not significant at 0.05

Complaint status is a leakage warning, not an early signal

Of 2,044 customers with Complain = 1, 2,034 exited (99.51%). The binary correlation between Complain and Exited is 0.9957. Complaint event timing is not available; the variable may be recorded at or after the churn event. It must not be presented as a valid pre-churn predictor or used to claim a causal churn trigger.

Statistical significance is not business causality. Interpret p-values alongside sample size, effect size, feature timing and the operational decision being considered.

6. Technical review: Python, SQL and model scope

Python EDA notebook

The supplied 52-cell notebook documents data loading, schema inspection, identifier removal, duplicate and missing-value checks, churn distribution, segment analyses, visualizations, chi-square tests, Mann-Whitney U tests, correlation analysis and an optional cleaned-data export. The notebook is EDA-focused and explicitly lists SQL, Power BI and machine-learning development as out of scope.

Important execution limitation: the uploaded notebook has no saved cell outputs or execution counts. Therefore, its saved state does not prove that the cells were executed in this submission. The headline metrics and statistical results in this report were independently recalculated with pandas and SciPy against the supplied raw CSV.

MySQL analysis script

The supplied MySQL 8+ script covers data preview, record/customer checks, key-field missingness, overall churn, geography, age, balance, product count, active membership, complaint status, satisfaction, card type, gender, combined activity/balance segments and a geography × product count × membership summary.

- The churn-rate denominator is correctly defined as the total records within each segment (COUNT(*)); NULLIF protects against division by zero.
- The script uses the original column names and backticks around the table name containing hyphens and around `Satisfaction Score`.
- No JOIN is present; all supplied queries operate on the single customer table, so join-key risk does not arise in this script.
- The script checks missingness for key columns rather than all 18 columns; the independent CSV scan found zero missing values across all 18.
- The script does not include an exact-row duplicate query; the raw CSV audit performed that check independently.

The SQL file was reviewed statically. No MySQL connection, imported table or query-result export was provided, so SQL execution and result equivalence remain unverified.

Predictive model claims

The separate business-problem brief contains legacy Logistic Regression and Random Forest metrics (including Random Forest recall 0.72 and ROC-AUC 0.87). However, the currently supplied EDA notebook explicitly excludes machine learning and contains no model-training or evaluation cells. These legacy scores were not re-run against the supplied CSV and are not presented as newly verified results. Do not describe this dataset as supporting a 30/60/90-day forecast: there are no event timestamps or defined lead-time labels.

7. Power BI dashboard review

The supplied screenshot shows a one-page dark dashboard with a KPI row and eight visual panels. The filter row now includes the four appropriate slicers: Geography, Gender, Card Type and Age Group. These are consistent with the project's categorical analysis and the accompanying DAX setup note.



Figure 1. User-supplied Power BI dashboard screenshot, reviewed 02 November 2026. Screenshot values are visually inspected; underlying PBIX/DAX execution was not supplied in this review.

Review item	Observation	Recommended correction
Slicers	Geography, Gender, Card Type and Age Group are present.	Keep all four; use dropdowns and provide a visible Reset Filters action.
KPI precision	Screenshot displays 10K, 2K, 20.38% and age 45.	Prefer 10,000; 2,038; 20.38%; and 44.84 years (or 44.8) for traceability.
Card-type visual	Bars are Diamond, Platinum, Silver and Gold but panel title says Number of Products.	Rename to 'Churn Rate by Card Type'.
Age-band order	Categories appear out of chronological order.	Sort Under 30, 30-39, 40-49, 50-59, 60+.
Balance / gender donuts	Center shows overall 20.38% while labels show category churn rates.	Use bar charts for rate comparisons or explicitly chart shares with a correct denominator.
Product-count panel	100% for four products and 82.71% for three products are shown.	Show sample counts (n=60 and n=266) prominently or in tooltips.
The screenshot's main geography, age, active-membership, balance, gender, satisfaction and product-count rates visually align with the CSV recalculation. The card-type panel title is mislabeled. This is a screenshot audit, not a live PBIX validation.		

8. Business recommendations and implementation roadmap

Priority	Recommended action	Evidence / rationale	Suggested owner
1	Run a Germany retention diagnostic.	Germany churn is 32.44% versus 16.17%-16.67% in France/Spain. Test whether the gap persists after comparing age, tenure, product count, balance and activity mix.	Retention + Regional Banking
2	Audit the 3+ product customer journey.	3-product churn is 82.71% (n=266); 4-product churn is 100% (n=60). Small groups and nonlinear pattern justify journey review, not a portfolio-wide policy change.	Product + Customer Experience
3	Pilot inactivity-based re-engagement.	Inactive customers show 26.87% churn versus 14.27% for active members. Use a controlled pilot and compare incremental retention.	CRM / Retention
4	Review balance and gender differences with context.	Has-balance and female segments show higher observed rates; inspect product mix, service history and cohort composition before attributing causes.	Customer Analytics + Risk
5	Strengthen event timing and model governance.	Complain is leakage-prone; no timestamps or lead-time label exist. Validate feature availability before any predictive use.	Data Governance + Model Risk

Proposed phased roadmap

Phase	Work	Exit criterion
A. Diagnose	Profile Germany, 3+ product and inactive segments; check customer mix and servicing journeys.	Segment facts and operational hypotheses reviewed with business owners.
B. Test	Design controlled re-engagement and focused segment pilots with a holdout/control group.	Treatment and control definitions agreed; baseline and outcome windows specified.
C. Validate	Collect event timestamps; rebuild leakage-aware features; validate on a later cohort if modeling is pursued.	Out-of-time metrics, calibration, fairness and threshold capacity reviewed.
D. Scale carefully	Compare incremental retention and customer value net of contact/incentive costs.	Scale decision based on controlled outcome and commercial evidence.

Ownership principle

The roles above are proposed functions, not named or formally assigned owners. The project files do not specify implementation dates, campaign capacity, contact costs, customer lifetime value or the bank's operating approval process.

9. Success measures, limitations and conclusion

Monitoring plan

Measure	How to evaluate
Incremental retention impact	Compare churn or retention outcomes in a campaign group with a comparable control group.
Targeting precision	Share of contacted customers who would otherwise churn, measured with a pre-defined outcome window.
Coverage / recall	Share of churners identified at the selected outreach capacity or score threshold.
Outreach efficiency	Churners identified per 100 outreach attempts; track reach and contactability.
Commercial value	Retained customer value less incentives, contact and servicing costs; requires finance-supplied values.
Model governance	Monitor calibration, drift, fairness and performance across geography and key customer groups.

Limitations and assumptions

- This is a historical cross-sectional customer dataset. It does not supply event dates, feature timestamps or an observation window suitable for time-based forecasting.
- Observed associations do not establish causal drivers. Root-cause claims require operational, product, service and customer-feedback evidence.
- The 3- and 4-product segments are small; the 100% rate for four products is based on 60 customers.
- Complaint status is almost identical to the outcome and has unknown timing; it is excluded from pre-churn interpretation.
- Card type and satisfaction score are not statistically associated with churn at the 0.05 level in these tests.
- The current Python notebook has no saved execution outputs; recalculated CSV results are presented separately from notebook execution status.
- The MySQL script was not executed against a database. The Power BI screenshot was visually inspected, but PBIX/DAX execution was not independently tested.
- Legacy model metrics in the business brief were not revalidated from the supplied EDA notebook and CSV during this run.

Evidence tracking

Claim / result	Source and calculation	Status
10,000 records; 2,038 exited; 20.38% churn	Raw CSV; count Exited and divide by 10,000.	Recalculated from source CSV
Geography, age, product, activity, balance, gender, card and satisfaction rates	Raw CSV; grouped counts, exited sums and within-group denominators.	Recalculated from source CSV
Statistical associations	SciPy chi-square and Mann-Whitney tests on raw CSV.	Recalculated from source CSV
SQL analysis logic	Supplied MySQL 8+ script; static review of query expressions.	Reviewed; not executed
Power BI labels and visual design	Supplied dashboard screenshot and DAX setup note.	Screenshot inspected; PBIX not executed
Historical model scores	Prior business-problem brief.	Legacy values; not revalidated

Conclusion

Prioritize a Germany diagnostic, a cautious review of the multi-product journey and controlled inactivity-based outreach. Do not operationalize prediction until feature timing, leakage controls and outcome windows are validated.

Bottom line: use the analysis to prioritize investigation, use controlled tests to establish impact, and use properly timed data before claiming predictive capability.